

# Towards Efficient and Privacy-Preserving Representation Learning on Dynamic Weighted Directed Graphs via Hashing Federated Latent Factorization of Tensors

## ‘Appendix’

Di Wu, *Member, IEEE*, Yaqian Zhan, Shuai Zhong, Yi He, *Member, IEEE*, Hongwei Li, *Fellow, IEEE* and Xin Luo, *Fellow, IEEE*

This is the Appendix File for the paper entitled “Towards Efficient and Privacy-Preserving Representation Learning on Dynamic Weighted Directed Graphs via Hashing Federated Latent Factorization of Tensors”. We have put some contents cited by the paper here.

### A. Symbols and Notations

In this study, a DWDG is formulated into a third-order HDI tensor, which is taken as a fundamental data source as shown in Fig. 1. To clarify, Table I provides a summary of the primary symbols and their respective meanings. Bold-face uppercase letters, bold lowercase letters, and lowercase letters are adopted to denote tensors, vectors, and scalars, respectively, throughout this paper.

TABLE I: Symbols and Notations.

| Symbol                                       | Explanation                                                                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{X}, \hat{\mathbf{X}}, \mathbf{X}_r$ | A three-order tensor for modeling a DWDG, Low-rank approximation to $\mathbf{X}$ , and Rank-one tensor regarding $\mathbf{X}$ .               |
| $I, J, K$                                    | $I$ and $J$ are two types of entity sets in a DWDG, and $K$ denotes the time slots set.                                                       |
| $x_{ijk}, \hat{x}_{ijk}, x_{ijk}^{(r)}$      | Single element in $\mathbf{X}, \hat{\mathbf{X}}$ , and $\mathbf{X}_r$ , respectively.                                                         |
| $R$                                          | Rank of $\hat{\mathbf{X}}$ and the dimension of latent feature space.                                                                         |
| $U, S, T$                                    | Latent feature matrices w.r.t. $I, J$ , and $K$ , respectively.                                                                               |
| $\mathbf{u}_i, \mathbf{s}_j, \mathbf{t}_k$   | The $i$ -th, $j$ -th, and $t$ -th row-vector of latent matrix $U$ for $i$ -th client, $S$ and $T$ .                                           |
| $\mathbf{u}_r, \mathbf{s}_r, \mathbf{t}_r$   | Latent feature vectors w.r.t. the $r$ -th rank-one tensor in $\hat{\mathbf{X}}$ ; also denote the $r$ -th column-vector of $U, S$ , and $T$ . |
| $u_{ir}, s_{jr}, t_{kr}$                     | Single latent feature in $\mathbf{u}_i, \mathbf{s}_j, \mathbf{t}_k$ .                                                                         |
| $\Lambda, \Gamma$                            | Known and Unknown entry set of $\mathbf{X}$ , respectively.                                                                                   |
| $D_i, D$                                     | The data collection of the $i$ -th data owner and all data from different data owners.                                                        |
| $M_{FED}, M_{SUM}$                           | The federated and centralized model, respectively.                                                                                            |
| $V_{FED}, V_{SUM}$                           | The accuracy of $M_{FED}$ and $M_{SUM}$ , respectively.                                                                                       |
| $\theta, N$                                  | The accuracy loss between $M_{FED}$ and $M_{SUM}$ . Maximum training iteration count.                                                         |
| $\lambda$                                    | The regularization coefficient of latent feature matrices.                                                                                    |

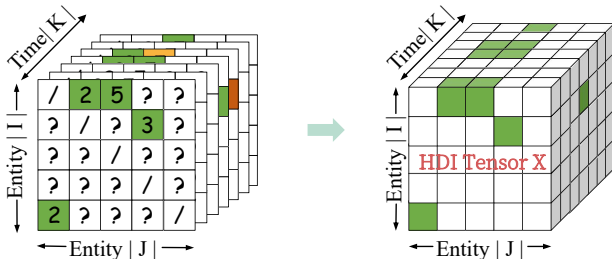


Fig. 1: An HDI tensor corresponding a DWDG.

**Definition 1 (An HDI tensor).** Given two types of entity sets  $I$  and  $J$ , and time slots set  $K$ ,  $\mathbf{X}^{|I| \times |J| \times |K|}$  is a three-order

TABLE II: Information of All the Datasets.

| Dataset         | CDs     | Network1 | Network2 | TV       |
|-----------------|---------|----------|----------|----------|
| Links           | 538997  | 2073902  | 1088913  | 1020731  |
| Sending nodes   | 23342   | 112600   | 14924    | 32005    |
| Receiving nodes | 18261   | 103347   | 18954    | 32927    |
| Timepoints      | 8       | 64       | 93       | 102      |
| Weight scale    | 1.0~5.0 | 1.0~20.0 | 0.0~24.0 | 1.0~5.0  |
| Sparsity        | 0.015%  | 0.0003%  | 0.004%   | 0.00094% |

tensor that quantizes certain relationships between  $I$  and  $J$  as time changes. Each element  $x_{ijk}$  of  $\mathbf{X}$  depicts a specific interaction between entity  $i \in I$  and entity  $j \in J$  at time slot  $k \in K$ . Two sets of  $\Lambda$  and  $\Gamma$  represent  $\mathbf{X}$ ’s known and unknown elements (i.e., the observed and unobserved links of a DWDG), respectively.  $\mathbf{X}$  is an HDI tensor because  $|I|$  and  $|J|$  are large and  $|\Lambda| \ll |\Gamma|$ .

### B. General Settings

**Datasets.** Four dataset CDs, Network1, Network2, and TV are as the benchmark. Table II summarizes their details. The 5-fold cross-validation is adopted in the experiments, i.e., every dataset is split up into five folds, where each fold is utilized as a testing set and the remaining parts are repeated as a training set. The ratio of the testing set is 20% of the dataset.

**Evaluation Metrics.** This research focuses on efficient representation learning on a DWDG. The link prediction of networks or missing data prediction tasks is typically devised as assessment protocols to depict a dynamic network [1], which directly demonstrates whether a model has effectively captured a DWDG’s vital characteristic. For this goal, root mean square error (RMSE) and mean absolute error (MAE) are adopted as common evaluation metrics [2]. They are calculated as follows:

$$RMSE = \sqrt{\sum_{x_{ijk} \in \Psi} (x_{ijk} - \hat{x}_{ijk})^2 / |\Psi|},$$

$$MAE = \sum_{x_{ijk} \in \Psi} |x_{ijk} - \hat{x}_{ijk}| / |\Psi|,$$
(1)

where  $\Psi$  represents the test set. A lower value of MAE and RMSE denotes a better prediction accuracy. To evaluate the efficiency of the proposed model, storage cost, communication volume, and inference time will be measured in the experiment.

**Baselines.** HF-LFT is compared with nine state-of-the-art models. There are eight real-valued models and a hashing-based model. The details of the nine models are depicted in Table III.

TABLE III: Summary of All the Involved Models.

| Model         | Description                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FPMF [3]      | The probabilistic matrix factorization-based federated recommendation system for rating prediction with explicit feedback. (IEEE Intelligent System 2021)                  |
| FSVD++ [3]    | The federated multilayered collaborative filtering framework with preserving privacy. (IEEE Intelligent System 2021)                                                       |
| MetaMF [4]    | The neural federated collaborative ranking is based on memory-based attention. (SIGIR 2020)                                                                                |
| FedRec++ [5]  | The lossless federated recommendation system depends on assigning denoising clients to decrease the impact of noisy information with the privacy-aware method. (AAAI 2021) |
| FedMF [6]     | A secure matrix factorization framework under the federated learning setting. (IEEE Intelligent System 2020)                                                               |
| PFedRec [7]   | A novel Personalized Federated Recommendation framework to learn many user-specific lightweight models to be deployed on smart device. (IJCAI 2023)                        |
| RFRRec [8]    | A efficient and robust regularized federated recommendation. (CIKM 2024)                                                                                                   |
| FedCIA [9]    | The information aggregation is achieved by leveraging collaborative information (especially item similarity) rather than directly employing model parameters. (SIGIR 2025) |
| LightFR [10]  | A lightweight federated recommendation framework with privacy-preserving matrix factorization. (TOIS 2023)                                                                 |
| <b>HF-LFT</b> | The proposed model in this paper.                                                                                                                                          |

**Implementation Details.** To conduct a fair comparison, the latent feature dimension is set to  $R=32$  for all the real-valued models and  $R=64$  for the hashing-based models. Setting the dimension in this manner is significant. The reason is that the goal is to save resources when compared to dense models while making link prediction performance as close to real-valued models as possible. Hence, the length of binary codes determines the threshold at which the HF-LFT becomes effective. Generally, the training round is set as 500 and adopt the early stopping technique. Besides, the length of the public key is specified as  $l = 1024$  in FedMF, the sampling parameter  $\rho = 3$  in FPMF, FSVD++, and FedRec++, the length of hidden layers  $L = 2$ , the number of hidden units  $h = 8$  and the length of low-dimensional node embeddings  $s = 8$  in MetaMF. All the hyperparameters are searched and tuned according to the performance of the test dataset. All datasets will perform five times of testing, and the average performance consequences will be adopted as the result.

### C. Accuracy Comparison with the basic LightFR (RQ.4)

The comparative results of HF-LFT and LightFR across four real-world datasets are illustrated in and Fig. 2 (MAE), from which two key phenomena can be observed. First, across all datasets and all binary bit length  $R = 8, 16, 32, 64, 128$ , HF-LFT consistently and significantly outperforms LightFR in terms of both MAE and RMSE. The main reason is that although HF-LFT and LightFR all utilize the L2H strategy, the former can capture the dynamic pattern of a DWDG. This verifies that capturing the dynamic characteristics is essential for a DWDG's representation learning. Second, both HF-LFT and LightFR exhibit an improvement in prediction accuracy as the binary bits increase, whereas HF-LFT is observably more stable and robust. The reason is that HF-LFT's dynamic modeling of DWDG provides a more robust and reliable representation foundation. This phenomenon further highlights the necessity of incorporating dynamic characteristics in DWDG representation learning.

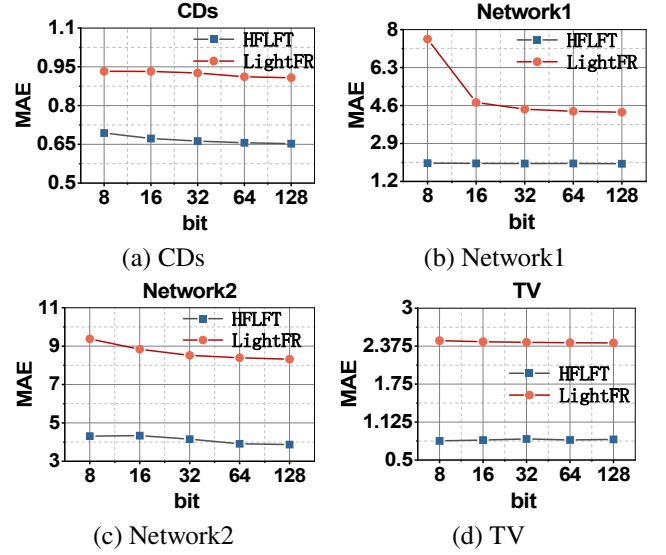


Fig. 2: The MAE of the HF-LFT and Light FR in different datasets, where  $R = 8, 16, 32, 64, 128$ .

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